Exp 12 - End-End Communication at Transport Layer

a) Implement echo client server using TCP/UDP sockets.

b) Implement a chat program using socket programming

TCP

```
import socket
import threading

#----- Server ----
def start_server():
    server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    server.bind(('127.0.0.1', 5000))
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000")

    conn, addr = server.accept()
    print(f"[SERVER] Connected by {addr}")
    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower() == "quit":
            print("[SERVER] Connection closed by the client.")
            break
        print(f"[SERVER] Received: {data}")
        conn.send(f"Server received: {data}".encode())
conn.close()
server.close
```

```
#------ CLIENT------
def start_client():
    client = socket.socket(socket.AF_INET, socket.SOCK_STEAM)
    client.connect(('127.0.0.1',5000)
    print("[CLIENT] Connected to server")

    While True:
        msg = input("Enter message (or 'quit to exit): ")
        client.send(msg.encode())
        if msg.lower()=='quit':
            break
        response = client.recv(1024).decode()
        print(f"[SERVER RESPONSE] {response}")
    client.close()

#---- MAIN----
if __name__=="__main__":
    #Run server in a background thread
    server_thread= threading.Thread(target=start_server,
daemon=True)
    server_thread.start()

    ## Run Run client is main thread
    start_client()
```

```
[SERVER] Listening on 127.0.0.1:5000
[CLIENT] Connected to server
[SERVER] Connected by ('127.0.0.1', 65267)
Enter message (or 'quit' to exit): hi
[SERVER] Received: hi
[SERVER RESPONSE] hi
Enter message (or 'quit' to exit): quit
[SERVER] Connection closed by client.
PS D:\sem 5\Computer Network\cn prac>
```

UDP

```
import socket
import threading

#-----SERVER------
def udp_server():
    s=socket.socket(socket.AF_INET,socket.SOCK_DGRAM)
    s.bind(('127.0.0.1',12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data,addr=s.recvfrom(1024)
        msg=data.decode()
        if msg.lower()=='quit':
        print(f"[SERVER] Client {addr} ended the connection.")
        break
        print(f"[SERVER] Received from {addr}:{msg}")
        s.sendto(data,addr) # Echo back
    s.close()
    print("[SERVER] Stopped.")
```

```
#----CLIENT----
def udp_client():
    c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message (or 'quit' to stop): ")
        c.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
            print("[CLIENT] Closing connection...") break
        data, _ = c.recvfrom(1024)
        print("Echo from server:", data.decode())
    c.close
    print("[CLIENT] Stopped.")

#----MAIN----
if __name__ == "__main__":
    threading.Thread(target = udp_server, daemon = True).start() udp_client
```

```
PS D:\sem 5\Computer Network\cn prac> python 12bserver.py
Server waiting for connection...
Connected with: ('127.0.0.1', 52099)
Client: hello
Server: hi
Client: how r u?
Server: i'm fine
Client ended the chat.
PS D:\sem 5\Computer Network\cn prac>
```